Two-Digit Multiplication with Area Models

Build the area model, estimate the product, find a missing partial product

Directions:

- Each rectangle is split by place value: the tens part and the ones part of one factor across the top, the tens part and the ones part of the other factor down the side.
- Write the product of the row label and the column label inside each region. Those four numbers are the *partial products*.
- Add the four partial products, then write the multiplication sentence on the answer line.

1. Multiply 23×14. The model is split for you.

	20	3
10		
4		

Answer: ________________

2. Multiply 32×15.

	30	2
10		
5		

Answer: ________________

3. Multiply 26×34. Both factors are large enough that every region matters — do not skip the two "cross" regions.

	20	6
30		
4		

Answer: ________________

4. **Estimate first.** Before drawing anything for 38×42, round each factor to the nearest ten and multiply the rounded factors. Write the estimate. (An estimate this close tells you roughly how big the whole rectangle will be.)

Answer: ____________________

5. Multiply 47×23.

	40	7
20		
3		

Answer: ____________________

6. This model for 54×26 has three regions filled in and one left empty. Fill the empty region, then find the product.

	50	4
20	1000	80
6	300	

Answer: ________________

7. Multiply 68×15.

	60	8
10		
5		

Answer: ________________

8. **Estimate.** Round each factor in 63×29 to the nearest ten, then multiply. Watch the second factor carefully — it is closer to one ten than to the other.

Answer: ________________

9. Multiply 45×37. This time *you* split the factors: write the tens part and the ones part of each factor in the blank labels, fill the four regions, and add.

	____	____
____		
____		

Answer: ____________

10. Maya multiplied 24×36 with an area model but wrote only two partial products: she multiplied the tens by the tens and the ones by the ones, and answered 624. Draw or complete the model below, say in one sentence which regions she left out, and give the correct product.

	20	4
30		
6		

Answer: ____________